Subcubic Storage Bounds beyond Scalar Recovery

Hans Brende GPT-6 Astra (OpenAI)

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Abstract

We extend the scalar storage bounds in *Curved Motion, Depth Reduction, and Storage Codes for Matrix Transposition on Tapes*. Reversible local relations over arbitrary alphabets admit a support-dependent generating set of size at most the matching number; a schedule uses each row at most once. This gives ambiguity and nullity payment bounds and recovers the original scalar theorems. Over any field, the intrinsic half bound holds at arbitrary coordinate widths when each recovery row has at most one component of rank above one. With unrestricted block ranks, the off-diagonal part of the squared recovery operator pays for excess kernel dimension. If that part vanishes, an exact quadratic rank identity holds. We also prove the required homomorphism theorem for square-covered subcubic graphs. All numbered results are verified in Lean. None of these additional coding hypotheses follows from native tape geometry alone.

Contributions. Hans Brende directed the research. GPT-6 Astra (OpenAI) developed the extensions, formalization, and text under his direction. The memo reuses the fourth paper's framework and graph lemmas, by Hans Brende and Claude Opus 5.5 (Anthropic).

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1 Local rules and the scalar theorem

The fourth paper proves a scalar linear storage bound on triangle-free subcubic graphs, including a more general directed-support form and an exact matching-number consequence [1]. A scalar row can solve for any coordinate with a nonzero coefficient. We isolate this property and retain the support combinatorics, without assuming scalar linearity. The resulting generating set applies to every alphabet and compatible code on the same supports.

Closure operators and their entropy are an established framework for network coding [2]. Our use of closure and polymatroids is within that framework. The result here is the matching-size generator bound for the specified local rules, together with a schedule that charges each center once. We do not claim a bound for arbitrary nonlinear recovery on all triangle-free subcubic graphs.

Sections 5 and 6 treat a different linear class: arbitrary coordinate widths with at most one higher-rank component per row. These bounds measure actual coordinate images and charge exceptional rows by their exact quotient deficit. Sections 7 and 8 treat arbitrary block ranks through the actual two-step defect. They give a weighted graph bound, a paid kernel bound, and an exact quadratic rank identity.

Every numbered result below has a Lean proof. Those declarations are the proofs; the English outlines are explanatory guides. The formalization imports the fourth paper's library and reuses its good-chain geometry and Tutte–Berge machinery. It proves new generating-set statements rather than assuming the old scalar dimension inequality for nonlinear objects.

Let V be finite. At each v choose $N_v \subseteq V \setminus \{v\}$ and write $R_v = \{v\} \cup N_v$. The general directed hypotheses are

$$p, q \in N_v \implies q \notin N_p, \tag{1.1}$$

$$|N_v| \le 2 \quad \text{or} \quad (|N_v| = 3 \text{ and } v \in N_u \implies u \in N_v \text{ for every } u). \tag{1.2}$$

They are exactly the no-recovery-triangle and reciprocity conditions used in the original directed theorem. In particular, they hold whenever $N_v \subseteq N_G(v)$ for a finite simple triangle-free graph G of maximum degree at most three. In the directed statements, G may be any simple undirected graph containing every pair vu with $u \in N_v$. Write $\nu(G)$ for its maximum matching size.

In Lean a matrix records the supports N_v. Only its zero pattern is used in the generator proof. Arbitrary named sets are encoded by a zero-one rational matrix; this introduces no field assumption on the recovered data.

2 A fixed generating set

Starting with known coordinates $B \subseteq W$, a rule centered at $v \in W$ may mark $x \in R_v \cap W$ known whenever all other members of $R_v \cap W$ are already known. A set generates W if finitely many such steps mark every member of W known. These are combinatorial rules; restricting a row to W does not assert that an arbitrary projected code can recover without its omitted coordinates.

Theorem 2.1 (Half-size generators and matching-size generators)**.** Lean: `exists_generators_half`, `exists_generators_matching` [`ReversibleRules.lean`]

Under Eqs. (1.1) and (1.2), every $W \subseteq V$ has a generating set $B \subseteq W$ with $|B| \le \lfloor |W|/2 \rfloor$. On $W = V$, if G contains the prescribed pairs, there is such a B with $|B| \le \nu(G)$. The choices depend only on the row supports.

Outline of the formal proof. Induct on $|W|$. If $|W|$ is positive and even, choose one seed and apply the odd-size induction bound to the remaining set, supplying the removed seed whenever a truncated

row needs it. For odd $|W|$, the original small-recovery lemma supplies a root with at most two parents in W. A good chain starts at this root. When a known center has two parents outside the chain, add both and seed one; its row derives the other. After j additions the chain has $2j+1$ vertices and at most $j+1$ seeds. Reciprocity ensures that every chain vertex has at most two parents outside.

If a chain row has just one outside parent, derive that parent without a seed. The enlarged set has at most half as many seeds as vertices; apply induction to its complement. If a pair remains, extend the chain. Finally, if no row has outside parents, either the singleton root derives itself without a seed, or the last pair shares a parent and has no prescribed edge between its members by Eq. (1.1). One member of that last pair is then derivable from the preceding chain, so its seed can be removed. Again the chain and complement meet the half-size bound. Finiteness terminates the chain construction.

For the matching bound, use the original library's Tutte–Berge partition: choose $U \subseteq V$ with $\nu(G) = |U| + \sum_A \lfloor |A|/2 \rfloor$, where A ranges over the components of $G - U$. Seed U and a half-size generating set in each component. Every omitted premise of a component row is in U, since no graph edge joins different components. Thus the component derivations lift to the whole vertex set, with at most $\nu(G)$ seeds. □

Theorem 2.2 (Productive schedule)**.** Lean: `Generates.exists_schedule`, `Schedule.centers_nodup`, `exists_matching_schedule` [`Schedules.lean`]

The set B in the matching conclusion can be accompanied by a finite list of steps (v_i, x_i), chosen from the supports, which marks only previously unknown targets and generates V. No center v_i occurs twice.

Outline of the formal proof. If a derivable target remains unknown, descend through its finite derivation until finding a rule all of whose other premises are already known. Apply that productive rule. The unknown set strictly shrinks. After using a row, all of its members are known, so that center can never again produce a new target. This last argument requires no graph hypothesis. □

A code $\mathcal{C} \subseteq \mathcal{A}^V$ is *reversible on these rows* if, for every v and every $x \in R_v$, agreement of two codewords on $R_v \setminus \{x\}$ implies agreement at x. Ordinary recovery of v from N_v is only one of these directions.

Theorem 2.3 (Alphabet-independent information set)**.** Lean: `reversible_information_set`, `named_rules_information_set`, `reversible_card_le`, `reversible_finrank_le` [`ReversibleRules.lean`]

There is a set B of size at most $\nu(G)$, determined by the supports, such that projection onto B is injective on every code reversible on the rows. In particular, for a finite nonempty alphabet of size q,

$$|\mathcal{C}| \le q^{\nu(G)}.$$

If the coordinates lie in one finite-dimensional vector space A over any field K, and $\mathcal{C} \le A^V$ is linear and reversible, then

$$\dim_K \mathcal{C} \le \nu(G) \dim_K A.$$

Outline of the formal proof. Two codewords agreeing on B agree at every step of a generating derivation. Projection onto B is therefore injective. Count the possible records in the finite-alphabet case. In the linear case this projection is an injective linear map into A^B, whose dimension is $|B| \dim_K A$. □

The set B is an information set, not an efficient native tape decoder. The theorem gives no bound on the cost of evaluating an individual inverse.

3 Linear-block consequences and the original theorem

Theorem 3.1 (Injective participating blocks)**.** Lean: `injective_block_information_set`, `injective_block_finrank_le` [`BlockRecovery.lean`]

Let G be finite, simple, triangle-free, and subcubic. Let A be a finite-dimensional vector space over any field K, and let $T_{vu} : A \to A$ be linear maps with $T_{vv} = I$ and $T_{vu} = 0$ off the edges for $v \neq u$. Suppose every nonzero T_{vu} is injective. For every linear subspace $\mathcal{C} \leq A^V$ satisfying $\sum_u T_{vu} x_u = 0$ at every v,

$$\dim_K \mathcal{C} \leq \nu(G) \dim_K A.$$

When A has positive dimension, the corresponding information set is chosen from the nonzero block pattern and works for every, possibly nonlinear, solution set. In finite dimension the nonzero injective endomorphisms are invertible; no commutation assumption is imposed.

Outline of the formal proof. Subtract two solutions agreeing at all but one participating member of a row. The remaining block applied to their target difference is zero; injectivity kills that difference. Thus every participating row is reversible, and Theorem 2.3 applies. The zero-dimensional case has zero code dimension directly. □

Corollary 3.2 (Original scalar statements)**.** Lean: `original_subCode_half`, `original_matching_bound` [`ReversibleRules.lean`]

Let M be a scalar matrix with unit diagonal, prescribed directed supports N_v, and Eqs. (1.1) and (1.2). Its original restricted code

$$\mathcal{C}_W = \{x : x_u = 0 \ (u \notin W), \quad (Mx)_v = 0 \ (v \in W)\}$$

satisfies $\dim \mathcal{C}_W \leq \lfloor |W|/2 \rfloor$ whenever Eq. (1.2) holds at the centers in W. If a graph G contains every prescribed pair, then $\dim \ker M \leq \nu(G)$.

Outline of the formal proof. A scalar row solves for each variable with nonzero coefficient. This remains true after outside coordinates are set to zero. Use the generating set of Theorem 2.1 and inject its restricted code into the scalar seed coordinates. The full-code matching statement follows from Theorem 2.3. Thus both statements are derived from the stronger rules theorem, not from a second copy of the original scalar induction. □

Corollary 3.3 (Exact scalar minrank)**.** Lean: `original_graph_matching`, `original_minrank` [`BlockRecovery.lean`]

For a triangle-free subcubic graph G, the minimum rank, over any field, of a scalar unit-diagonal matrix with off-diagonal support on G is $|V| - \nu(G)$.

Outline of the formal proof. The preceding nullity bound and rank–nullity give the lower bound. Reuse the original library's matching-supported construction for attainment, and apply the new upper nullity bound to get equality. This is a rederivation of the original minrank result, not a new minrank value. □

4 Paying for ambiguous inversions

The same support-only schedule remains useful when some directions of a row are not exactly reversible. In this section the supports satisfy Eqs. (1.1) and (1.2) and fit G; B and (v_i, x_i) are the fixed seed set and schedule from Theorem 2.2.

For a finite code $\mathcal{C}$ and an attained row pattern, let $L_{v,x}$ bound the number of possible values at $x \in R_v$ after all other coordinates of R_v are fixed. These are inverse-fibre bounds on the actual code, not bounds on the number of whole codewords in a fibre.

Theorem 4.1 (Finite ambiguity ledger)**.** Lean: `paid_cardinality_schedule`, `paid_cardinality_matching` [`CardinalityCosts.lean`]

For any finite code over any alphabet,

$$|\mathcal{C}| \le |\pi_B \mathcal{C}| \prod_i L_{v_i,x_i}.$$

If the alphabet is finite and nonempty with size q, and each row has a common bound $L_v \ge 1$ valid for all its directions, then

$$|\mathcal{C}| \le q^{\nu(G)} \prod_{v \in V} L_v.$$

Outline of the formal proof. At a productive step, all other coordinates in its row have already been observed. Each previous observation therefore has at most L_{v_i,x_i} extensions when x_i is added. Multiply these finite fibre bounds along the schedule. At the end every codeword is distinguished. For the second bound, $|\pi_B \mathcal{C}| \le q^{|B|}$, and the centers are distinct, so replacing directional costs by $L_v \ge 1$ charges each row at most once. Empty codes cause no exception. □

A real polymatroid is a function $h : 2^V \to \mathbb{R}$ with $h(\varnothing) = 0$, monotonicity, and submodularity. No integrality or exchange axiom is assumed. For every $x \in R_v$ set $\delta_{v,x} = h(R_v) - h(R_v \setminus \{x\})$.

Theorem 4.2 (Polymatroid ambiguity ledger)**.** Lean: `Schedule.polymatroid_directions`, `paid_polymatroid_matching`, `reversible_polymatroid_matching` [`PolymatroidBounds.lean`]

For every real polymatroid h,

$$h(V) \le h(B) + \sum_i \delta_{v_i,x_i}.$$

If $h(\{v\}) \le 1$ and $0 \le \delta_{v,x} \le \kappa_v$ for every participating direction, then

$$h(V) \le \nu(G) + \sum_v \kappa_v.$$

In particular, if all the rows are reversible for h, meaning $h(R_v) = h(R_v \setminus \{x\})$ for every $x \in R_v$, then $h(V) \le \nu(G)$.

Outline of the formal proof. Submodularity bounds the marginal of an added target after all known coordinates by its marginal after only the other row coordinates. Sum these inequalities along the schedule. For the row-budget form, use $h(B) \le \sum_{v \in B} h(\{v\}) \le |B| \le \nu(G)$ and charge each center once. Zero directional costs give the last statement. □

The original library proves that finite joint entropy is such a polymatroid, for every source law. The exact schedule formula therefore applies to conditional-entropy costs as well. The normalized singleton bound is needed only for the displayed $\nu(G)$ specialization.

Theorem 4.3 (Singular-block nullity ledger)**.** Lean: `Schedule.nullity_directions`, `paid_block_matching` [`BlockCosts.lean`]

Let A be a finite-dimensional vector space over any field K, and let $T_{vu} : A \to A$ be linear maps vanishing for $u \notin R_v$. Suppose $\mathcal{C} \le A^V$ satisfies $\sum_u T_{vu} x_u = 0$ for every v. Then the fixed schedule gives

$$\dim_K \mathcal{C} \le |B| \dim_K A + \sum_i \dim_K \ker T_{v_i x_i}.$$

Consequently, if $\dim_K \ker T_{vx} \le \kappa_v$ *for every* $x \in R_v$,

$$\dim_K \mathcal{C} \le \nu(G) \dim_K A + \sum_v \kappa_v.$$

No invertibility, common eigenbasis, or commutation premise is required.

Outline of the formal proof. Let Z_S be the subspace of codewords vanishing on the known set S. If a rule at v is ready to add x, its row equation gives $T_{vx} f_x = 0$ for every $f \in Z_S$. Evaluation at x maps Z_S into $\ker T_{vx}$, with kernel $Z_{S\cup\{x\}}$. Rank–nullity yields $\dim Z_S \le \dim \ker T_{vx} + \dim Z_{S\cup\{x\}}$. Telescope along the schedule, whose final vanishing subspace is zero. Projection onto the seeds gives $\dim \mathcal{C} \le |B| \dim A + \dim Z_B$. The row-maxima version again uses distinct centers. The general restriction lemmas are shared in `LeanMachine.FiniteProjection`. □

For an above-matching code, these inequalities require enough directional ambiguity on every such schedule. They do not assert that a native tape computation has small ambiguity or small block nullities.

5 Rank-one ports with arbitrary coordinate widths

The reversible-rule bounds require an input to be recoverable from the other symbols in its row. A singular coefficient need not permit this. We now allow arbitrary finite coordinate widths, provided every recovery block has rank at most one. Throughout this section, K is any field and $G = (V, E)$ is a finite simple triangle-free graph of maximum degree three.

Theorem 5.1 (R: intrinsic half bound)**.** *Let* A_v *be finite-dimensional* K*-vector spaces and let* $C \le \prod_{v\in V} A_v$ *be a linear code. Suppose linear maps* $T_{vu} : A_u \to A_v$ *satisfy*

$$x_v = \sum_{u\in V} T_{vu} x_u \quad (x \in C), \qquad T_{vu} = 0 \text{ if } vu \notin E, \qquad \operatorname{rank} T_{vu} \le 1.$$

Writing π_v *for coordinate projection, we have*

$$2 \dim C \le \sum_{v\in V} \dim \pi_v(C) \le \sum_{v\in V} \dim A_v.$$

Consequently, if B *is the block operator with blocks* T_{vu} *and* $N = \sum_v \dim A_v$*, then* $2 \operatorname{rank}(I - B) \ge N$.
Lean: `LowRank.rank_one_code_half`, `LowRank.rank_one_code_capacity`, `LowRank.rank_one_block_rank`, `LowRank.scalar_recovery_half` [`RankOneCodes.lean`]

Outline of the formal proof. The proof first establishes a statement about an arrangement of subspaces $U_v \le W$ spanning W. Its supplied component maps $R_{vu} : U_v \to W$ have images in U_u, vanish off G, have rank at most one, and sum to the inclusion of U_v in W. It proves $2 \dim W \le \sum_v \dim U_v$.

Induct on $\dim W$. Quotienting by any nonzero subspace S produces another arrangement of the same kind: choose a linear section of each local quotient image and compose each old component with that section and the ambient quotient. This constructs new maps of no larger rank; it does not assume that the old maps descend. A counterexample would therefore satisfy

$$f(S) := 2 \dim S - \sum_v \dim(U_v \cap S) \ge 1 \qquad (S \ne 0).$$

Distinct coordinate spaces are disjoint. Each coordinate has dimension at most two: its at most three image lines span it, and equality with a three-dimensional row span would contradict that disjointness.

The submodularity of f makes intersecting spaces of deficit one have deficit-one intersection and sum. Equality also gives the exact identity $U_v \cap (S + T) = (U_v \cap S) + (U_v \cap T)$ for those spaces. A plane coordinate's three input lines span a three-dimensional space of deficit one, meeting no other coordinates. Two adjacent plane rows would have a tight intersection supported on just two lines, an impossible odd deficit. Thus every neighbor of a plane is a line.

Starting with one plane's row span, grow a tight space by either adjoining two outside lines whose nonzero row contributions sum into the space, or absorbing a plane's row span along one shared input line. Exact intersection accounting preserves whole-coordinate membership and the inclusion of every known plane's row span. Every known line retains a known neighbor. A reachable space of maximum dimension is closed under all the supplied component maps: a partially met plane span would admit an absorption, and a known line has at most two outside neighbors, whose nonzero contributions would admit a line extension.

If this closed space is proper, its coordinate intersections span it and the actual recovery maps restrict to it. Its deficit one contradicts induction. If it is the whole space, inspect the last growth operation. An initial star or a final line extension contradicts recovery of one of its leaves. For a final star absorption, the two newly added input axes are independent modulo the predecessor space. Triangle-freeness and this independence make that predecessor closed under all the old rows, again contradicting induction. If there was no plane to begin with, evaluation on line generators gives a scalar code; the scalar half bound applies, counting only nonzero coordinates.

For the stated code, set $W = C^*$ and $U_v = \operatorname{im} \pi_v^*$. Local linear sections of π_v^* turn the dual recovery equations into the required component maps without increasing their ranks. These coordinate duals span C^*, and $\dim U_v = \dim \pi_v(C)$. The ambient-capacity bound follows by inclusion, and rank–nullity gives the block-operator statement. Width one recovers the scalar graph half bound. □

The arrangement theorem and its global closed-subspace step are
`Lean`: `LowRank.rank_one_arrangement_half`,
`LowRank.Core.exists_smaller_closed` [`RankOneBound.lean`]
The generic quotient, restriction, and code-dual constructions are shared in the repository's `lean-machine` library. In particular, no compatible choice of a scalar basis across different ports is assumed. For the construction program, a code exceeding half capacity must escape the rank-one condition somewhere. This is a restriction on the recovery maps; it is not a condition supplied by tape geometry.

Theorem 5.2 (Paid exceptional rows)**.** *Consider a spanning arrangement $U_v \le W$ with supplied recoveries as above, but allow unrestricted component ranks at vertices in $D \subseteq V$. Every row outside D still has rank-at-most-one components. Set $H = \sum_{v \in D} U_v$ and $\Delta = 2 \dim W - \sum_v \dim U_v$. Then*

$$\Delta \le 2 \dim H - \sum_v \dim(U_v \cap H) \le \dim H \le \sum_{v \in D} \dim U_v.$$

`Lean`: `LowRank.rank_one_exception_defect`,
`LowRank.exceptional_span_defect` [`RankOneCodes.lean`]

Outline of the formal proof. Quotient by H, using the local-section construction. Exceptional coordinates become zero, and all surviving components have rank at most one. Apply the arrangement half bound to this quotient. The identities $\dim(U_v \bmod H) = \dim U_v - \dim(U_v \cap H)$ give the first inequality. The sum of the exceptional coordinate dimensions is at least $\dim H$ and contributes in full to the intersection sum, giving the remaining inequalities. □

Thus any excess above half must be paid by the actual span of exceptional coordinates. The theorem does not assert that this span is small in a native tape computation.

6 One higher-rank component in each row

The rank-one condition can be relaxed at one port in every row, without bounding that port's rank or any original coordinate width. The graph hypotheses and intrinsic capacity convention remain those of Section 5.

Theorem 6.1 (H: one higher-rank port)**.** *Let $C \le \prod_v A_v$ have prescribed linear recoveries $x_v = \sum_u T_{vu} x_u$, with $T_{vu} = 0$ off a finite simple triangle-free subcubic graph. Suppose, for each v, at most one map T_{vu} has rank greater than one. Then*

$$2 \dim C \le \sum_v \dim \pi_v(C) \le \sum_v \dim A_v.$$

For the corresponding block operator B of total scalar size N, this gives $2 \operatorname{rank}(I - B) \ge N$.
Lean: `LowRank.one_high_code_half`, `LowRank.one_high_code_capacity`, `LowRank.one_high_block_rank` [`HigherRankCodes.lean`]

Outline of the formal proof. Use the coordinate-dual arrangement and induction on ambient dimension, as in Theorem 5.1. Local sections preserve the one-higher-rank condition in every quotient. A counterexample would again have $f(S) \ge 1$ for every nonzero test space, and distinct coordinate spaces would be disjoint.

Each coordinate is a line or plane. For a row with a higher-rank component R_{vu}, the remainder $x \mapsto x - R_{vu}x$ is injective: a vector in its kernel would lie in both U_v and U_u. Its image is spanned by the at most two remaining rank-one ports, so $\dim U_v \le 2$. The previous three-line argument handles rows without a higher-rank port. Thus every higher-rank component in this hypothetical core is a rank-two isomorphism between planes. Call its directed edge a high arc.

A high arc $v \to u$ forces a high successor $u \to w$ with $w \ne v$. Otherwise $U_v + U_u$ has dimension four and contains the two endpoint planes and four exterior input lines at distinct vertices. Their eight coordinate dimensions contradict positive deficit. This argument includes a low-rank or zero reverse port; reciprocity is not assumed.

Along a high path $p \to a \to b \to c$, let z and y be the third graph neighbors of a and b. The space

$$S = U_a + U_b + \operatorname{im} R_{by}$$

has dimension at most five and contains the planes U_a, U_b, U_c, as well as input lines at p, z, y. Triangle-freeness makes these six vertices distinct. Their total contribution is nine, forcing $\dim S = 5$, $f(S) = 1$, and exactly those coordinate intersections.

The next successor plane lies in S plus the next third-neighbor line. That line is outside S. Adding it grows the space by one dimension and increases its own coordinate intersection. Positive deficit forces the successor plane to have already met S in a line. The exact support just proved then forces either a high four-cycle or an edge from a to that successor.

A high four-cycle is impossible. Starting from its first tight triple space, adjoining the next third-neighbor line grows dimension by one. It gains one dimension at the remaining cycle plane and at each of two distinct external coordinates. If an external vertex coincides with one from the previous triple, the neighboring triple's exact intersection shows that its new line is independent of the old line. Thus there are three coordinate gains for two units of dimension budget, contradicting positive deficit.

Seven consecutive high arcs now suffice for the final contradiction. The forced distance-three edges and maximum degree three make the last vertex equal the second. The six intervening distinct vertices form $K_{3,3}$, with every available graph edge used. Recovery makes all six plane coordinates lie in the span of one three-vertex part. That span has dimension at most six but at least twelve coordinate-intersection dimensions, again contradicting positive deficit. Every high arc could be extended to such a path, so there are none. The rank-one arrangement bound finishes the induction. The code and block-rank conclusions use the same dualization and rank–nullity as before. □

The arbitrary-width arrangement theorem is
Lean: `LowRank.one_high_arrangement_half` [`HigherRankBound.lean`]

Its two final exclusions are
Lean: `LowRank.HighCore.no_high_four_cycle` [`HigherRankFourCycle.lean`]
Lean: `LowRank.HighCore.no_seven_path`, `LowRank.HighCore.no_high` [`HigherRankExclusion.lean`]

The proof does not require a classification of the full high-arc graph. In particular, it does not assume that high arcs have distinct predecessors.

Corollary 6.2 (Exceptional rows for Theorem H)**.** *In the arrangement version of Theorem 6.1, allow rows at $D \subseteq V$ to have arbitrarily many higher-rank components. With $H = \sum_{v \in D} U_v$ and $\Delta = 2 \dim W - \sum_v \dim U_v$, the exact payment remains*

$$\Delta \le 2 \dim H - \sum_v \dim(U_v \cap H) \le \dim H \le \sum_{v \in D} \dim U_v.$$

Lean: `LowRank.one_high_exception_defect` [`HigherRankCodes.lean`]
Lean: `LowRank.exceptional_span_defect` [`RankOneCodes.lean`]

Outline of the formal proof. Quotient by H. Exceptional coordinates become zero; all other rows retain at most one higher-rank component. Apply the arrangement theorem and the same exact intersection accounting as in Theorem 5.2. □

For a construction exceeding half capacity, some rows must have at least two higher-rank ports, and the span of those exceptional coordinates must pay the excess. This does not imply a matching-number bound for arbitrary vector codes, nor a small exceptional span for native computations. The reversible-rule and one-higher-rank hypotheses are separate sufficient conditions; neither is asserted for unrestricted linear or nonlinear recovery.

7 Graphs covered by four-cycles

The next bounds impose no restriction on individual block ranks. Their graph input concerns the edges in the *original* support graph that belong to a simple four-cycle. Write $G_\square$ for this spanning subgraph, and $F = E(G) \setminus E(G_\square)$. There is no iterative edge-deletion process in this definition.

Theorem 7.1 (C: a common target for square-covered graphs)**.** *Let G be a finite simple triangle-free graph of maximum degree three. If every edge belongs to a four-cycle, then G admits a graph homomorphism to the circular clique $K_{8/3}$. Consequently every normalized, monotone, submodular function $h : 2^V \to \mathbb{R}$ satisfying $h(N_G(v) \cup \{v\}) = h(N_G(v))$ obeys*

$$2h(V) \le \sum_{v \in V} h(\{v\}).$$

In particular, $G_\square$ satisfies this weighted half bound for every triangle-free subcubic G, whether or not all edges of G lie in squares.

Lean: `Squares.target`, `Squares.good`, `squareCore_good` [`SquareGood.lean`]

Outline of the formal proof. Work first in a connected component. Two small configurations force a closed bipartite component: two vertices with three common neighbors, and an embedded cube with one vertex deleted. The formal proof treats both by the same boundary lemma. A bipartite seed has three boundary vertices, each already adjacent to two internal vertices. A square through any new edge forces its outside endpoint to touch two boundary vertices. Any two such pairs intersect, and the degree bound then forces all outside attachments to be the same vertex. The enlarged component remains bipartite and maps to an edge of $K_{8/3}$.

Suppose now that no such homomorphism exists. The preceding configurations are absent. Distinct vertices have at most two common neighbors, and no vertex can support squares on all three pairs of incident edges: those three squares would produce a cube cap, or three common neighbors.

An embedded ladder consists of disjoint two-vertex rungs joined by two rails. A square with an outside attachment extends to a three-rung ladder; if no attachment exists, the square is a closed bipartite component. Choose a longest embedded ladder, of at least three rungs. Each interior vertex already has degree three. At an endpoint, a new edge must belong to a square through the endpoint's rung. A square through the last rail instead would give three common neighbors or three square wedges at the preceding interior vertex. The new square's other two vertices both lie outside the ladder, by degree saturation. It would therefore extend the ladder, a contradiction.

Connectedness makes this ladder span the component. Its extra edges can only form a partial matching between its two end rungs. Complete that matching: the resulting graph is a closed ladder, with either parallel or twisted closing rails. Adding two rungs admits a fold onto the shorter ladder, by inserting a two-edge backtrack. Four finite base maps suffice: the three-rung twisted ladder and four-rung untwisted ladder are bipartite; the four-rung twisted ladder is the Möbius ladder M_8; the five-rung untwisted ladder maps to its five-cycle. The three-rung untwisted closure is excluded by triangle-freeness. The Lean kernel checks these finite maps directly.

Label M_8 by $\mathbb{Z}/8\mathbb{Z}$, with edge differences $1, -1, 4$. Multiplication by three identifies it with $K_{8/3}$, whose edge differences are $3, 4, 5$. Combine the component maps. Finally apply the fourth paper's formalized homomorphism pullback and its certified weighted half bound for $K_{8/3}$ [1]. Every edge of $G_\square$ retains its original witnessing square, so the same theorem applies to that subgraph. □

For the construction program, squares alone cannot provide a subcubic support exceeding half capacity, even for arbitrary polymatroid recovery. The next section accounts for the edges outside those squares without assuming that deleting them preserves an arbitrary code.

8 Two-step defects and exact quadratic ranks

Let K be any field, let A_v be finite-dimensional K-vector spaces, and put $A = \prod_v A_v$ and $N = \sum_v \dim A_v$. Fix a finite simple triangle-free subcubic graph G and an operator $W : A \to A$ with blocks $T_{vu} : A_u \to A_v$ vanishing off its edges. In particular $T_{vv} = 0$. Define

$$E = \operatorname{offdiag}(W^2), \qquad E_{ac} = \begin{cases} \sum_b T_{ab} T_{bc}, & a \neq c, \\ 0, & a = c. \end{cases}$$

All ranks below are ordinary K-linear ranks, including when K has characteristic two.

Call (a, c) touched if $a \neq c$ and a two-edge path $a - b - c$ contains an edge of $F = E(G) \setminus E(G_\square)$. Such a path has a unique middle vertex; a second one would put both edges in a square. Define the source-separated constraint map

$$\Phi : A \longrightarrow \prod_{c \in V} A, \qquad (\Phi x)_{c,a} = \begin{cases} E_{ac} x_c, & (a, c) \text{ touched}, \\ 0, & \text{otherwise}. \end{cases}$$

Unlike Ex, this map does not sum contributions from different source coordinates. Set $C = \ker(I - W)$, $H = C \cap \ker \Phi$, and $q = \operatorname{rank}(\Phi|_C) = \dim C - \dim H$.

Theorem 8.1 (S: paid two-step stability)**.** *With the preceding definitions,*

$$\begin{aligned} 2 \dim H &\leq \sum_v \dim \pi_v(H), \\ N &\leq 2 \operatorname{rank}(I - W) + 2q, \qquad q \leq \operatorname{rank} \Phi \leq 6 \operatorname{rank} E, \\ N &\leq 2 \operatorname{rank}(I - W) + 12 \operatorname{rank} E, \\ N &\leq 2 \operatorname{rank}(I + W) + 12 \operatorname{rank} E. \end{aligned}$$

Lean: `subcubic_constrained_half`, `subcubic_paid_rank`, `subcubic_twoStep_stability` [`QuadraticBounds.lean`]

Lean: `constraintLoss_identity`, `constraintLoss_le` [`TwoStepStability.lean`]

Lean: `separatedConstraint_rank_le` [`TwoStepDefect.lean`]

Outline of the formal proof. For an edge $uv \in F$, define $p_{uv}x$ to have coordinates $T_{uv}x_v$ at u, $T_{vu}x_u$ at v, and zero elsewhere. On the *full* space H, the two-step constraints imply that this vector again belongs to H, recovers across uv, and transmits zero along every other incident edge. Thus p_{uv} is an idempotent on H; the idempotents for distinct unordered edges annihilate one another. Each endpoint projection is injective on its edge factor, giving exactly twice the factor dimension in total coordinate capacity.

This decomposition also splits capacity coordinate by coordinate. At u the operator p_{uv} is induced by $T_{uv}T_{vu}$ on $\pi_u(H)$, and similarly at v; at every other coordinate it is zero. Hence each coordinate image is the direct sum of its edge-factor images and its image from the residual $\ker \sum_{uv \in F} p_{uv}$. On that residual all off-square transmissions vanish. Its actual coordinate observations therefore recover on $G_\square$. Apply Theorem 7.1 to their rank polymatroid, then add back the balanced edge factors. This proves the intrinsic bound for H. No invariance is asserted for an arbitrary subcode of C.

Rank–nullity gives $q + \dim H = \dim C$ and the paid bound. To compare Φ with E, each block row of E has at most six possible source coordinates: three first neighbors, each with at most two nonreturning continuations. Choose a basis from the *actual* scalar row functionals of E. Each chosen row has support in at most six blocks, so the sum of the block-column ranks is at most $6 \operatorname{rank} E$. Projecting each column onto its touched targets cannot increase rank, and source separation makes these ranks add. Therefore $\operatorname{rank} \Phi \leq 6 \operatorname{rank} E$. Replacing W by $-W$ leaves E unchanged and gives the final inequality. □

The actual-kernel restriction and coordinatewise splitting are checked in

Lean: `edgeProjection_idempotent`, `edgeProjection_annihilate`, `edgeProjection_coordinatewise`, `edgeProjection_balanced` [`EdgeProjections.lean`]

For a full kernel of dimension $(1/2 + \varepsilon)N$, the theorem requires $q \geq \varepsilon N$ and $\operatorname{rank} E \geq \varepsilon N/6$. This rules out a small two-step defect as the source of a fixed excess over half; it does not prove that an arbitrary native recovery operator has small defect.

Theorem 8.2 (Q: exact quadratic rank identity)**.** *If $E = 0$, equivalently W^2 is block diagonal, then*

$$2\,\mathrm{rank}(I - W) = 2\,\mathrm{rank}(I + W) = N + \mathrm{rank}(W^2 - I).$$

Lean: `subcubic_quadratic_exact` [`QuadraticBounds.lean`]

Outline of the formal proof. First suppose $W^2 = I$. The zero-defect case of Theorem 8.1 gives $\mathrm{rank}(I \pm W) \geq N/2$. The identity $(I - W)(I + W) = 0$ gives the reverse inequality for their sum. Both ranks, and both kernel dimensions, are therefore exactly $N/2$, without an eigenspace decomposition or a restriction on characteristic.

In general write $W^2 = \mathrm{diag}(D_v)$ and set $S_v = \ker(D_v - I)$. Then $S = \ker(W^2 - I) = \prod_v S_v$. Commutation of W with W^2 shows that every actual block T_{vu} maps S_u into S_v. The restricted operator on S is an involution on the same graph support. Both $\ker(I - W)$ and $\ker(I + W)$ lie in S, so their dimensions equal those of the restricted kernels, namely $\dim S/2$. Finally $\dim S = N - \mathrm{rank}(W^2 - I)$ and rank–nullity give the displayed equalities. □

Corollary 8.3 (Injective-subspace refinement and involutions)**.** *Under $E = 0$, if $W^2 - I$ is injective on any subspace $U \leq A$, then*

$$N + \dim U \leq 2\,\mathrm{rank}(I - W), \qquad N + \dim U \leq 2\,\mathrm{rank}(I + W).$$

If $W^2 = I$, both doubled ranks equal N.
Lean: `subcubic_quadratic_injective`, `subcubic_involution_ranks` [`QuadraticBounds.lean`]

Outline of the formal proof. Injectivity gives $\dim U \leq \mathrm{rank}(W^2 - I)$; apply Theorem 8.2. If $W^2 = I$, that rank is zero. □

The injective-subspace statement includes any invariant summand on which $W^2 - I$ is invertible, but needs neither invariance nor a chosen Fitting decomposition. The exact identity is stronger: it accounts for the whole rank of $W^2 - I$. For a construction, block-diagonal two-step behavior cannot support a kernel above half ambient capacity, regardless of widths, commutativity, or singular individual blocks. These are additional sufficient hypotheses for the half bound; they do not imply the original scalar matching bound for unrestricted vector codes.

9 Construction requirements and remaining questions

These bounds distinguish solving a row for its center from solving it for any participating coordinate. A construction on the prescribed supports that exceeds the matching bound must violate the latter property. The paid versions quantify the required failure: inverse fibres, conditional marginals, or singular-block kernels must contribute enough along the fixed schedule. Invertible changes of alphabet at individual coordinates do not remove an information-set bound.

For linear codes, exceeding half the total coordinate-image dimension also requires some rows with at least two components of rank greater than one. The span of those exceptional coordinates must pay the excess. This criterion permits arbitrarily large original coordinate widths and does not require compatible scalar bases at different ports.

Independently, the squared recovery operator must have enough off-diagonal rank: a full kernel of dimension $(1/2 + \varepsilon)N$ requires $\mathrm{rank}\,\mathrm{offdiag}(W^2) \geq \varepsilon N/6$. The exact loss on the kernel can be smaller than this global rank payment. The quadratic identity further excludes every block-diagonal square, over all characteristics. Square-covered support graphs are already subject to a weighted

polymatroid half bound, so the remaining possibility must use off-square edges with nonzero two-step transmission.

The main transposition question (T) concerns a fixed finite-tape machine running in $O(L)$ time, where $L = d^2 b(d)$ and $b(d) = \max(1, \lceil \log_2 d \rceil)$. Nothing here supplies the necessary subcubic representation for arbitrary tape computations, proves local reversibility there, or pays the cost of implementing the inverse maps. Those are separate construction or impossibility obligations. Even on a triangle-free subcubic graph, arbitrary nonlinear recovery and unrestricted singular-block recovery remain outside the unqualified half bound. The next useful construction would simultaneously evade the reversible, one-higher-rank, and small-defect criteria while meeting native tape costs. A lower-bound approach must instead show why an adequate restriction or payment is forced by the actual computation.

A Index of Lean declarations

Declarations are relative to `SubcubicMemo`; their axiom checks are generated together with this index (Appendix B).

SubcubicMemo.BlockCosts `Schedule.nullity_directions`, `paid_block_matching`.

SubcubicMemo.BlockRecovery `injective_block_finrank_le`, `injective_block_information_set`, `original_graph_matching`, `original_minrank`.

SubcubicMemo.CardinalityCosts `paid_cardinality_matching`, `paid_cardinality_schedule`.

SubcubicMemo.EdgeProjections `edgeProjection_annihilate`, `edgeProjection_balanced`, `edgeProjection_coordinatewise`, `edgeProjection_idempotent`.

SubcubicMemo.HigherRankBound `LowRank.one_high_arrangement_half`.

SubcubicMemo.HigherRankCodes `LowRank.one_high_block_rank`, `LowRank.one_high_code_capacity`, `LowRank.one_high_code_half`, `LowRank.one_high_exception_defect`.

SubcubicMemo.HigherRankExclusion `LowRank.HighCore.no_high`, `LowRank.HighCore.no_seven_path`.

SubcubicMemo.HigherRankFourCycle `LowRank.HighCore.no_high_four_cycle`.

SubcubicMemo.PolymatroidBounds `Schedule.polymatroid_directions`, `paid_polymatroid_matching`, `reversible_polymatroid_matching`.

SubcubicMemo.QuadraticBounds `subcubic_constrained_half`, `subcubic_involution_ranks`, `subcubic_paid_rank`, `subcubic_quadratic_exact`, `subcubic_quadratic_injective`, `subcubic_twoStep_stability`.

SubcubicMemo.RankOneBound `LowRank.Core.exists_smaller_closed`, `LowRank.rank_one_arrangement_half`.

SubcubicMemo.RankOneCodes `LowRank.exceptional_span_defect`, `LowRank.rank_one_block_rank`, `LowRank.rank_one_code_capacity`, `LowRank.rank_one_code_half`, `LowRank.rank_one_exception_defect`, `LowRank.scalar_recovery_half`.

SubcubicMemo.ReversibleRules `exists_generators_half`, `exists_generators_matching`, `named_rules_information_set`, `original_matching_bound`, `original_subCode_half`, `reversible_card_le`, `reversible_finrank_le`, `reversible_information_set`.

SubcubicMemo.Schedules `Generates.exists_schedule`, `Schedule.centers_nodup`, `exists_matching_schedule`.

SubcubicMemo.SquareGood `Squares.good`, `Squares.target`, `squareCore_good`.

SubcubicMemo.TwoStepDefect `separatedConstraint_rank_le`.

`SubcubicMemo.TwoStepStability` `constraintLoss_identity`, `constraintLoss_le`.

B Building and verification

Clone the complete repository with sibling directories intact. The memo uses Lean and Mathlib v4.34.1 and imports the fourth paper's package through `../../lean`, including its dependencies on earlier and shared libraries. In this memo's `lean/` directory run

```
lake exe cache get
lake build
lake env lean PaperAxioms.lean
```

From the memo directory, `python3 tools/lean_index.py` generates both the declaration index and the axiom checks from the LaTeX references. All cited declarations depend at most on `propext`, `Classical.choice`, and `Quot.sound`. No new axioms, `sorry`, or kernel-bypassing evaluation are used. The original paper's sources are unchanged. Build the PDF in `paper/` with `latexmk -pdf main.tex`.

The square-covered graph theorem is proved in the library, including the closed-component argument, ladder extension, finite base maps, and component assembly. It is not supplied as an assumption to the final rank theorems. The shared library supplies sparse-row rank accounting, coordinatewise idempotent splitting, and invariant-kernel transport. One routine elaboration, the constrained-kernel capacity sum, raises `maxHeartbeats` to 1,200,000; this changes no logical rule.

References

[1] Hans Brende and Claude Opus 5.5 (Anthropic). Curved motion, depth reduction, and storage codes for matrix transposition on tapes. Manuscript and partial Lean formalization, repository HansBrende/papers, 2026. Pinned revision b6aaab17c9458e9ca39b35bd29a060910b71fdbe. URL: https://github.com/HansBrende/papers/tree/b6aaab17c9458e9ca39b35bd29a060910b71fdbe/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes.

[2] Maximilien Gadouleau. Entropy of closure operators and network coding solvability. *Entropy*, 16(9):5122–5143, 2014. URL: https://www.mdpi.com/1099-4300/16/9/5122, doi:10.3390/e16095122.